

1. Budget-set draw

$T = 25$ rounds, $K = 2$ goods
continuous-density prices
fresh seed per replicate

2. Prompt / format arm

baseline, reciprocal,
multiturn; 3 arms at 1.5B,
2 arms at 3B

3. Parse + capped retry

$\geq 20/25$ valid rounds:
keep. Else retry, capped
at 3 attempts per slot

residual discard: 8 of 150 slots

5. MILP projection

Demuynck & Rehbeck (2023) minimal-quantity-
error repair, re-verified by the same check

4. GARP check

combinatorial Warshall-closure
check; 85 of 142 traces violate

GARP-consistent (57 of 142): no projection

6a. Real repaired sequence

$\tilde{x}$: GARP-consistent by
construction, independently
verified; dose = its L_1 distance

6b. Matched-null construction

GARP-blind: shrink every bundle toward the
fixed center, at an identical L_1 displacement
primary (information-fair) null; oracle null

7. Payoff scoring, under both payoff designs

original: fixed equal-weight Cobb-Douglas, one optimum at $s = 0.5$ for every trace
corrected: per-trace $\alpha_s \sim \text{Uniform}(0.05, 0.95)$, read only at scoring time
 Δpayoff , real repair vs. each null — paired, at matched displacement